

Academic Qualifications

Year	Degree/Certificate	Institute	CPI/%
2024 - Present	B.Tech	Indian Institute of Technology Kanpur	8.7/10
2024	JKBOSE(XII)	Burn Hall School, Sonwar, Srinagar	93.6%
2022	JKBOSE(X)	Burn Hall School, Sonwar, Srinagar	96.6%

Work Experience

Founder's Office Intern - Product and Growth Cherry Date   (Dec'25 - Feb'26)			
Objective	● Spearheaded the zero-to-one market entry for a safety-centric dating platform to redefine the local ecosystem.		
Approach	● Drove Women First user acquisition through safety propositions of Time Compensation & Ironclad Safety .		
Impact	● Established an initial market footprint by converting offline engagement into a verified early-adopter pipeline.		

Research Experience

Depression Classification and Prediction Prof. Hamim Zafar ML - Intern Cosmic Lab  (May'26 - Present)			
Objective	● Build a multimodal ML pipeline for Major Depressive Disorder classification & HAMD prediction from MRI.		
Approach	● Processed 12,000+ NifTI scans of 2,380 subjects from 25 sites into 6,670 connectivity & 580 ROI features/sub. ● Compared Linear SVM , Random Forest , XGBoost , and a multimodal GNN for depression prediction.		
Result	● Achieved ROC-AUC 0.68 with multimodal GNN fusion superseding the state-of-the-art in MDD prediction.		

Key Projects

Deepfake Detection EE-656 (Course Project) Prof. Nischal Kumar Verma IIT Kanpur  (May'26 - Present)			
Objective	● Benchmark deepfake detection systems to assess accuracy, efficiency, and robustness under media degradations.		
Approach	● Trained MesoNet & Xception on 100K+ images; evaluated AUC-ROC , FLOPs , robustness & Grad-CAM .		
Results	● Achieved AUC-ROC of 0.97/0.99 and identified MesoNet for edge deployment and Xception for robustness.		

Predictive Market Modelling Association of Computing Activities IIT Kanpur  (Jun'26 - Present)			
Objective	● Develop a hybrid ARIMA-LSTM framework combining ARIMA trend extraction with LSTM residual modeling.		
Approach	● Combined ADF/ACF/PACF-driven ARIMA trend modeling pipeline with LSTM residual forecasting .		
Results	● Achieved 4.31% (Nifty) and 6.69% (TCS) test MAPE, outperforming standalone ARIMA & LSTM baselines.		

Brain spy: Deciphering Alzheimer's Brain and Cognitive Science Club IIT Kanpur  (May'25 - Sep'25)			
Objective	● Leverage 3D structural MRI and used attention-based explainability for automated disease classification.		
Approach	● Developed ThreeHeadADNet , a multi-view dual-CNN integrating axial , coronal , and sagittal MRI views. ● Designed a modified AXIAL explainable CNN on a frozen ResNet152 backbone with Attention-XAI Fusion .		
Result	● Got 97% accuracy (F1 = 0.965), corrected a data-leakage flaw & won the Best Research Project Award .		

MediQ - Medical AI Chatbot Society of Civil Engineers  (Dec'25 - Jan'26)			
Objective	● To engineer a full-stack ML web app delivering real-time, explainable symptom-based disease diagnosis.		
Approach	● Built a Flask REST API over pickled artifacts , paired with a JS dashboard & symptom autocomplete .		
Result	● Built a dashboard with confidence gauge , top-5 differentials & severity scoring at sub-second latency .		

QuantOption: Multi-Model Option Pricing Platform Self Project  (May'26 - Jun'26)			
Objective	● Build an end-to-end option pricing platform integrating Machine Learning and quantitative pricing models .		
Approach	● Combined Black-Scholes , Monte Carlo , Binomial Tree , and XGBoost/RF with Greeks and backtesting.		
Results	● Benchmarked 5 pricing models on AAPL options and deployed a real-time interactive Streamlit dashboard.		

Relevant Courses & Technical Skills (*) Ongoing

Fundamentals of Computing	Probability & Statistics	Linear Algebra, ODE, PDE	AI, ML, DL & Applications*
● Programming & Softwares: C/C++, Python, HTML, CSS, $\text{\LaTeX}$, SQL, CAD Fusion 360, Simulink, Docker, Ansys, Arduino			
● Libraries & Framework: OpenCV, NumPy, Pandas, Matplotlib, TensorFlow, Scikit-Learn, PyTorch, Nibabel, Seaborn, Nilearn			

Positions of Responsibility & Extra-Curriculars

- Coordinated **Roadtrip Junoon** as **Organiser**, Events & Competitions, **Antaragni'26**, managing a budget of over **INR 2 Cr+**. 
- Secured over **INR 7 Lakh+** in sponsorships, collaborated with **50+ bands**, and conducted **Road Trips** across **7+ metro cities**.
- Elected by a **600+ member electorate** as the **Maintenance Secretary** of Hall III managing a budget of **INR 15 lack** rupees.
- Serving as an **Ex-Officio Member** of the **Anti-Ragging Committee**, contributing to student safety and **grievance handling**.
- Led **11 techincal problem statements** simultaneously as **Hall Captain** during the **Intra-IIT Technological Competition**.